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SUBJECT CODE NO: - RNP-8012
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group B) CSE / AI & ML / EEE (NEP) Sem - II
Examination June/July 2025
Basics of Electronics Engineering

[Time: 3 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from Part B: Q.2 to Q.7.
3. Use of non-programmable calculator is allowed

Part A**Q1 Solve the following questions****20**

1. Define Zener Diode & LED with symbols
2. Match the pairs:

Sr. No.	Value of CFR Resistor	Color Codes
1	5.2 M Ω with 20% Tolerance	Yellow, Violet, Red, Brown
2	120 K Ω with 10% Tolerance	Green, Red, Green
3	4.7K Ω with 1% Tolerance	Brown, Black, Gold, Gold
4	1 Ω with 5% Tolerance	Brown, Red, Yellow, Silver

3. Match the pairs:

Sr. No.	Device	Properties
1	Transducer	Current Controlled
2	MOSFET	Integrator
3	BJT	Voltage Controlled
4	Op-Amp	Measurement of physical quantity

4. In a PN junction Diode the potential barrier is due to the charges on either side of the junction, these charges are:
 - a) Majority carriers
 - b) Minority carriers
 - c) Both (a) and (b)
 - d) Positive & negative ions
5. Op-Amp can amplify:
 - a) Only AC signal
 - b) Both (a) and (b)
 - c) Only DC signal
 - d) Only Noise signal
6. Draw the circuit diagrams of Integrator & Differentiator

7. Define CMRR & Slew Rate of OP-Amp, state its importance
8. Convert following Decimal number into its equivalent Binary number:
 $(44.625)_{10} = (?)_2$
9. Convert following Binary number into its equivalent Decimal number:
 $(100.111)^2 = (?)_{10}$
10. Define HLR & VLR in Mobile Communication System

Part B

Attempt ANY FOUR questions from Part B: Q.2 to Q.7.

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|-----------|---|-----------|
| Q2 | (a) Draw & Explain the Block diagram of DC Regulated Power Supply | 05 |
| | (b) What is a Rectifier? State its types & explain any one with suitable diagrams | 05 |
| Q3 | (a) Define FET with types, structure & symbol. Explain its working | 05 |
| | (b) Define MOSFET with types, structure & symbol. Explain its working | 05 |
| Q4 | (a) Draw & Explain the Block diagram of Op-Amp | 05 |
| | (b) What is a summing Amplifier? Explain with suitable diagrams | 05 |
| Q5 | (a) What are different Number Systems with suitable examples | 05 |
| | (b) Explain all Basic Logic Gates with symbols & Truth Tables | 05 |
| Q6 | (a) Define a Transducers, state its classification | 05 |
| | (b) Explain the working of Thermocouple. State its advantages & disadvantages | 05 |
| Q7 | (a) Explain the elements of a Communication System with block diagram | 05 |
| | (b) What is Modulation? State its types & Need of Modulation | 05 |